

LUXEUIL AB, France

WMO: 072920

Lat: 47.787N

Lon: 6.364E

Elev: 278

StdP: 98.03

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-9.3	-6.6	-12.3	1.3	-7.3	-9.2	1.8	-4.6	11.5	8.3	10.4	7.6	1.2	60	0.498

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.2	31.3	20.6	29.4	20.0	27.7	19.4	21.8	28.7	20.9	27.5	20.1	26.1	2.6	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.7	14.9	24.4	18.7	14.0	23.4	17.9	13.3	22.5	64.9	28.8	61.6	27.6	58.7	26.3	26.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.0	7.5	6.2	DB	-13.2	33.9	3.4	2.2	-15.6	35.5	-17.6	36.8	-19.5	38.0	-22.0	39.6	(4)
(5)			WB	-13.4	23.7	3.2	0.9	-15.7	24.4	-17.6	24.9	-19.5	25.5	-21.8	26.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.5	2.1	3.2	6.6	9.9	13.8	17.5	19.2	18.8	14.8	11.2	5.9	2.7
	DBStd	7.14	4.20	4.03	3.67	3.62	3.50	3.76	3.15	3.35	3.24	3.72	3.95	4.23
	HDD10.0	999	246	191	116	46	7	1	0	0	2	31	132	227
	HDD18.3	3007	504	424	364	253	144	61	27	35	113	222	374	485
	CDD10.0	1184	1	1	10	43	127	224	284	273	146	68	7	1
	CDD18.3	152	0	0	0	0	6	35	53	50	8	1	0	0
	CDH23.3	1893	0	0	0	19	108	441	651	566	101	8	0	0
	CDH26.7	587	0	0	0	1	15	135	226	193	17	0	0	0
Wind	WSAvg	2.4	2.6	2.7	2.7	2.5	2.3	2.2	2.2	2.0	2.0	2.2	2.4	2.6
Precipitation	PrecAvg	1216	105	97	86	82	105	94	95	93	96	114	123	126
	PrecMax	1689	217	191	275	187	165	188	203	209	203	227	273	261
	PrecMin	893	26	23	9	16	25	26	28	7	22	15	24	12
	PrecStd	171	53	50	62	47	34	42	41	47	50	55	67	64
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.7	16.1	20.9	25.7	29.0	33.0	34.2	33.8	28.9	24.5	17.9	13.5
		MCWB	10.0	9.5	11.8	15.2	19.2	21.1	20.9	21.6	19.0	17.1	13.0	10.6
	2%	DB	11.2	13.4	18.6	22.9	26.3	30.1	31.2	30.9	26.4	21.8	14.9	11.6
		MCWB	9.3	8.7	11.2	14.1	17.6	20.3	20.4	20.7	19.1	16.5	12.1	9.7
	5%	DB	9.8	11.2	16.1	20.4	24.3	28.1	29.2	28.8	24.0	19.4	13.2	10.2
		MCWB	8.3	8.0	10.0	12.9	16.4	19.6	19.8	20.1	18.0	15.3	11.1	8.8
	10%	DB	8.1	9.4	13.5	18.0	22.1	25.9	27.2	26.7	21.9	17.2	11.6	8.9
		MCWB	7.0	7.2	9.0	11.6	15.3	18.8	19.1	19.5	16.6	14.3	9.9	7.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.1	10.7	13.2	16.2	20.1	22.8	22.7	22.9	21.9	18.4	13.9	11.3
		MCDB	12.0	13.5	18.7	24.0	26.9	29.9	29.9	30.2	25.4	22.3	16.1	12.4
	2%	WB	9.6	9.5	11.8	14.6	18.6	21.3	21.5	21.7	19.6	17.0	12.6	10.1
		MCDB	10.9	12.4	16.4	21.0	24.9	28.0	28.8	28.7	24.3	20.7	14.4	11.2
	5%	WB	8.3	8.4	10.7	13.4	17.1	20.2	20.6	20.8	18.4	15.8	11.2	9.0
		MCDB	9.6	10.7	14.8	19.2	22.6	26.5	27.5	27.1	23.0	18.7	12.9	10.2
	10%	WB	7.0	7.3	9.7	12.2	15.9	19.2	19.7	19.9	17.2	14.6	10.1	7.7
		MCDB	8.1	9.3	13.0	17.1	20.8	25.1	25.7	25.7	21.0	17.0	11.4	8.8
Mean Daily Temperature Range	5% DB	MDBR	6.4	8.3	10.7	12.1	11.8	12.3	12.2	12.3	11.6	10.1	7.3	6.3
		MCDBR	7.9	12.8	16.5	17.3	16.3	15.8	16.5	16.5	15.3	13.3	10.2	7.8
		MCWBR	6.1	8.4	9.7	9.3	8.3	7.3	7.2	7.5	8.5	8.2	7.2	6.0
	5% WB	MCDBR	7.1	10.3	13.6	15.3	14.3	14.1	14.7	14.7	13.2	12.0	9.2	7.2
		MCWBR	5.8	7.3	8.3	8.5	7.6	6.9	7.0	7.1	7.9	7.8	6.9	5.8
Clear-Sky Solar Irradiance	taub	0.311	0.333	0.362	0.396	0.403	0.409	0.405	0.401	0.377	0.361	0.329	0.307	
	taud	2.446	2.390	2.330	2.252	2.273	2.291	2.298	2.325	2.371	2.405	2.446	2.445	
	Ebn at Noon	768	824	853	854	861	855	853	840	827	779	735	724	
	Edh at Noon	66	86	106	127	129	127	125	117	101	84	66	59	
All-Sky Solar Radiation	RadAvg	1.02	1.82	3.06	4.33	5.07	5.93	5.62	4.88	3.71	2.28	1.19	0.83	
	RadStd	0.14	0.30	0.47	0.63	0.53	0.53	0.51	0.47	0.36	0.20	0.14	0.16	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (79 neighbors)	N/A	N/A	N/A	N/A	+0.37	+0.26	N/A	N/A	N/A	N/A

Nomenclature: See separate page